ECE250: Signals & Systems Quiz-8

Date: 7th November, 2025 **Total Marks:** 10 **Duration:** 15 Minutes

---

Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[CO1 CO2 CO4 CO5] [Q1]: Consider a causal LTI system described by the difference equation

$$y[n] + \frac{1}{2}y[n-1] = x[n].$$

(a) Determine the frequency response $H(e^{j\omega})$ of this system. **[3 Marks]**

(b) What is the response of the system to the following input- **[3 Marks]**

$$x[n] = \left(-\frac{1}{2}\right)^n u[n]$$

(c) Find the response to the inputs with the following Fourier transform- **[4 Marks]**

$$X(e^{j\omega}) = 1 + 2e^{-3j\omega}$$